

RECORD ME

Every maths trick in one sitting — 92 passages

Read this file top to bottom into any voice recorder (phone Voice Memos is perfect). One long recording is fine — or split into a few takes wherever suits, as long as each take starts at a numbered item.

The three rules

1. Before each passage, say its number clearly — "Number one", "Number two" — then pause for a second, read the passage, and pause for a couple of seconds before the next number. Those pauses are how I find the joins. 2. If you stumble, don't stop the recording: just pause, say the SAME number again ("Number twelve"), and read that passage again. I'll keep your retake. 3. Read like you're explaining at the kitchen table — warm and a bit energetic, not newsreader-flat. Grins are audible!

When you're done: save the file(s) (m4a, mp3 or wav — anything) into a new folder called recordings inside your OneDrive Times tables folder, then tell me. I'll split it by your spoken numbers, attach each piece to its trick in the game (whiteboards will play YOUR voice), and publish.

Estimated reading time: 45–60 minutes. A cuppa is advised.

Tick the box next to each number as you record it, so you can pause and pick up where you left off.

Part A — Times-table grid tricks

■ Say: "Number 1" — Double, double

To multiply by four, you double the number, then double it again. Let's try seven times four. Double seven is fourteen. Double fourteen is twenty-eight. So seven times four is twenty-eight. Two easy doubles instead of one hard multiply.

■ Say: "Number 2" — Times ten, then halve

To multiply by five, times the number by ten, then cut it in half. Let's try eight times five. Eight times ten is eighty. Half of eighty is forty. So eight times five is forty. Times ten, then halve — every single time.

■ Say: "Number 3" — Times five, plus one more lot

To multiply by six, do times five first, then add one more lot of the number. Let's try seven times six. Seven times five is thirty-five. Add one more seven: forty-two. So seven times six is forty-two.

■ Say: "Number 4" — Times five, plus times two

Seven is just five plus two — so times seven is times five plus times two. Let's try eight times seven. Eight times five is forty. Eight times two is sixteen. Forty plus sixteen is fifty-six. So eight times seven is fifty-six.

■ Say: "Number 5" — Double three times

To multiply by eight, double the number three times in a row. Let's try six times eight. Double six is twelve. Double twelve is twenty-four. Double twenty-four is forty-eight. So six times eight is forty-eight. Three quick doubles.

■ Say: "Number 6" — Nine fingers

Here's the famous finger trick for the nines. Hold up all ten fingers, and bend down the finger you're multiplying by. For seven times nine, bend down finger number seven. Now look: six fingers on the left of the bent one, three on the right. Six and three — sixty-three! Seven times nine is sixty-three. Your hands know the nine times table.

■ Say: "Number 7" — Times ten, minus one lot

To multiply by nine, do times ten, then take one lot of the number away. Let's try six times nine. Six times ten is sixty. Take away one six: fifty-four. So six times nine is fifty-four.

■ **Say: "Number 8" — Times ten**

Times ten is the easiest trick in the book: just put a zero on the end. Seven times ten? Seventy. Twenty-three times ten? Two hundred and thirty. Every digit slides up one place and a zero fills the gap.

■ **Say: "Number 9" — Times eleven, single digit**

To multiply a single digit by eleven, just write the digit twice. Seven times eleven is seventy-seven. Four times eleven is forty-four. The number simply looks in the mirror.

■ **Say: "Number 10" — Times eleven, two digits**

For a two-digit number times eleven, here's the magic: add the two digits together, and slot the answer in between them. Let's try thirty-five times eleven. Three plus five is eight. Slot the eight between the three and the five: three, eight, five. Thirty-five times eleven is three hundred and eighty-five. If the digits add to ten or more, carry the one into the front digit.

■ **Say: "Number 11" — Times ten plus times two**

Twelve is ten plus two, so times twelve is times ten plus times two. Let's try six times twelve. Six times ten is sixty. Six times two is twelve. Sixty plus twelve is seventy-two. So six times twelve is seventy-two.

■ **Say: "Number 12" — Times twenty-five**

Twenty-five is a quarter of a hundred. So to times by twenty-five, times by a hundred and then divide by four. Let's try eight times twenty-five. Eight times a hundred is eight hundred. Divide by four: two hundred. So eight times twenty-five is two hundred. Think of it like money — eight twenty-five pence coins make two pounds.

■ **Say: "Number 13" — Times fifty**

Fifty is half of a hundred. So times fifty means times a hundred, then halve it. Let's try six times fifty. Six times a hundred is six hundred. Half of that is three hundred. So six times fifty is three hundred.

■ **Say: "Number 14" — Times ninety-nine**

Ninety-nine is just one less than a hundred. So times ninety-nine means times a hundred, minus one lot of the number. Let's try seven times ninety-nine. Seven times a hundred is seven hundred. Take away one seven: six hundred and ninety-three. So seven times ninety-nine is six hundred and ninety-three.

■ **Say: "Number 15" — Times a hundred**

Times a hundred: put two zeros on the end. Seven times a hundred is seven hundred. Forty-two times a hundred is four thousand two hundred. Every digit jumps up two places.

■ **Say: "Number 16" — Round to a tidy number, then adjust**

When a number is close to something tidy, use the tidy number and fix it after. Let's try six times twenty-nine. Twenty-nine is nearly thirty. Six times thirty is one hundred and eighty. But we used one extra lot of six, so take six back: one hundred and seventy-four. So six times twenty-nine is one hundred and seventy-four.

■ **Say: "Number 17" — Split into tens**

When you multiply by a round number like forty, split it: times four first, then add the zero. Let's try seven times forty. Seven times four is twenty-eight. Add the zero: two hundred and eighty. So seven times forty is two hundred and eighty.

■ **Say: "Number 18" — Split by place value**

For a big number, break it into tens and ones, multiply each bit, and add them up. Let's try six times twenty-three. Split twenty-three into twenty and three. Six times twenty is one hundred and twenty. Six times three is eighteen. One hundred and twenty plus eighteen is one hundred and thirty-eight. So six times twenty-three is one hundred and thirty-eight.

■ **Say: "Number 19" — Squaring a number ending in five**

Any number ending in five is easy to square. Take the front part, multiply it by one more than itself, then stick twenty-five on the end. Let's try thirty-five squared. The front is three. Three times four is twelve. Stick twenty-five on the end: one thousand two hundred and twenty-five. Thirty-five squared is one thousand two hundred and twenty-five. Like magic — but it's maths.

■ **Say: "Number 20" — Squaring a round ten**

To square a round number like seventy, square the front digit and stick two zeros on. Seven squared is forty-nine, so seventy squared is four thousand nine hundred.

■ **Say: "Number 21" — Square via the next round number**

To square a number just under a tidy one, use this: the square equals one less times one more, plus one. Let's try squaring twenty-nine. Twenty-eight times thirty would work, but even easier: twenty-nine squared is thirty times twenty-eight, plus one. Thirty times twenty-eight is eight hundred and forty. Add one: eight hundred and forty-one. Twenty-nine squared is eight hundred and forty-one.

■ **Say: "Number 22" — Both numbers end in five**

When both numbers end in five, like forty-five times twenty-five: take the front digits — four and two. Multiply them: eight. Add their average — three — to get eleven... actually here's the simple version: multiply the fronts, add half the sum of the fronts, and finish with twenty-five. Forty-five times twenty-five: four times two is eight, plus half of four plus two — that's three — gives eleven, then twenty-five on the end: one thousand one hundred and twenty-five. Check it — it works!

■ **Say: "Number 23" — Teens times teens**

For two teen numbers, like thirteen times fourteen: add the ones digits to one number, times ten, then add the product of the ones. Thirteen plus four is seventeen. Seventeen times ten is one hundred and seventy. Three times four is twelve. One hundred and seventy plus twelve is one hundred and eighty-two. So thirteen times fourteen is one hundred and eighty-two.

■ **Say: "Number 24" — Same tens digit**

When both numbers share the same tens digit, like twenty-three times twenty-six, use twenty as your base. Twenty-three plus six is twenty-nine. Twenty-nine times twenty is five hundred and eighty. Then three times six is eighteen. Five hundred and eighty plus eighteen is five hundred and ninety-eight. So twenty-three times twenty-six is five hundred and ninety-eight.

■ **Say: "Number 25" — Same tens, ones add to ten**

Here's a beauty. If the tens digits match and the ones digits add to ten — like forty-three times forty-seven — multiply the tens digit by one more than itself: four times five is twenty. Then multiply the ones: three times seven is twenty-one. Write them together: two thousand and twenty-one... that is, twenty then twenty-one: forty-three times forty-seven is two thousand and twenty-one. Instant!

■ **Say: "Number 26" — Tens add to ten, same ones**

The twin of the last trick. If the TENS add to ten and the ONES match — like forty-six times sixty-six — multiply the tens and add one lot of the ones digit: four times six is twenty-four, plus six is thirty. Then square the ones: six times six is thirty-six. Write them together: thirty, thirty-six — three thousand and thirty-six. Forty-six times sixty-six is three thousand and thirty-six.

■ **Say: "Number 27" — Close to one hundred**

For two numbers just under a hundred, like ninety-seven times ninety-four: see how far each is from a hundred. Ninety-seven is three away. Ninety-four is six away. Cross-subtract: ninety-seven minus six is ninety-one — that's the front of your answer. Multiply the gaps: three times six is eighteen — that's the back. Ninety-one, eighteen: nine thousand one hundred and eighteen. Ninety-seven times ninety-four is nine thousand one hundred and eighteen. An ancient Indian trick called Nikhilam.

■ **Say: "Number 28" — Close to ten**

The same cross trick works near ten. Take eight times seven. Eight is two away from ten, seven is three away. Cross-subtract: eight minus three is five — the tens. Multiply the gaps: two times three is six — the ones. Fifty-six! Eight times seven is fifty-six.

■ **Say: "Number 29" — Square the middle, minus the gap squared**

When two numbers sit evenly around a nice middle — like eighteen times twenty-two, sitting around twenty — square the middle and subtract the square of the gap. Twenty squared is four hundred. The gap is two, and two squared is four. Four hundred minus four is three hundred and ninety-six. So eighteen times twenty-two is three hundred and ninety-six.

■ **Say: "Number 30" — One step from a square**

If you know your squares, neighbours are free. Seven times eight is just seven squared plus one more seven: forty-nine plus seven is fifty-six. Or eight squared minus eight: sixty-four minus eight — fifty-six again. Any square you know unlocks the numbers either side of it.

■ **Say: "Number 31" — Halve and double**

If one number is even, you can halve it and double the other — the answer stays the same. Let's try sixteen times twenty-five. Halve sixteen: eight. Double twenty-five: fifty. Eight times fifty is four hundred. Do it again if you like: four times a hundred — four hundred. Same answer, much friendlier numbers.

■ **Say: "Number 32" — Regroup the factors**

Sometimes it helps to break a number into friendlier factors. Let's try eighteen times five. Eighteen is nine times two. So that's nine times two times five — and two times five is ten. Nine times ten is ninety. So eighteen times five is ninety. Look for pairs that make ten or a hundred — they do the work for you.

■ **Say: "Number 33" — The two-hand trick for six to ten**

Number your fingers from six to ten on each hand, thumbs being six. To multiply seven times eight, touch the seven finger to the eight finger. Count the touching fingers and everything below: that's five — fifty. Multiply the fingers above on each hand: three times two is six. Fifty plus six: fifty-six. Seven times eight is fifty-six — and it's always in your hands.

Part B — Addition tricks

■ **Say: "Number 34" — Left to Right**

Left to Right means you add the BIGGEST parts first. Take $34 + 52$. Split each number into tens and ones. Add the tens first: $30 + 50 = 80$. Then add the ones: $4 + 2 = 6$. Put them together: $80 + 6 = 86$. Always start with the tens!

■ **Say: "Number 35" — Number Bonds**

Number bonds are pairs of numbers that make 10. $7 + 3 = 10$. $6 + 4 = 10$. $8 + 2 = 10$. $5 + 5 = 10$. Learn them so you never have to count them out. They work for 100 too: $70 + 30$, $60 + 40$, $80 + 20$. Knowing your bonds makes every other trick faster.

■ **Say: "Number 36" — Bigger Number First**

Adding is easier if you start with the BIGGER number. For $3 + 28$, do not start at 3. Flip it round: $28 + 3$. The answer is exactly the same. Now just count on from 28: 29, 30, 31. So $3 + 28 = 31$.

■ **Say: "Number 37" — Count On**

When you add 1, 2 or 3, do not work it out — just count up. $46 + 3$: start at 46... 47, 48, 49. Three little hops. $46 + 3 = 49$. It works counting on from any number.

■ **Say: "Number 38" — Doubles**

A double is a number added to itself. $6 + 6 = 12$. $7 + 7 = 14$. $8 + 8 = 16$. $9 + 9 = 18$. Learn them off by heart — then they are instant. Doubles are the anchors that lots of other sums lean on.

■ **Say: "Number 39" — Near Doubles**

A near double is ALMOST a double. $6 + 7$ is nearly $6 + 6$. Double the 6 gives 12, then add the extra 1 gives 13. So $6 + 7 = 13$. Tip: for numbers 2 apart (like $7 + 9$), double the MIDDLE one — double $8 = 16$.

■ **Say: "Number 40" — Make Ten**

Make Ten means jumping to the next ten first. $8 + 5$: how many gets 8 up to 10? That is 2. Split the 5 into $2 + 3$. $8 + 2 = 10$, then $10 + 3 = 13$. So $8 + 5 = 13$.

■ **Say: "Number 41" — Spot the Tens**

When you add lots of numbers, look for pairs that make 10. Take $6 + 3 + 4 + 7$. Spot them: $6 + 4 = 10$, and $3 + 7 = 10$. Add the tens: $10 + 10 = 20$. So $6 + 3 + 4 + 7 = 20$.

■ **Say: "Number 42" — Round and Adjust**

Round one number UP to something tidy, then fix it afterwards. $38 + 27$: round 38 up to 40 — that is 2 extra. Now the easy sum: $40 + 27 = 67$. Take back the 2 you added: $67 \text{ minus } 2 = 65$. So $38 + 27 = 65$.

■ **Say: "Number 43" — Give and Take**

Give and Take keeps the total exactly the same. $47 + 26$: give 3 from the 26 across to the 47. The 47 becomes 50. The 26 becomes 23. Now it is easy: $50 + 23 = 73$. So $47 + 26 = 73$.

■ **Say: "Number 44" — Add 9 and 99**

Adding 9 is nearly the same as adding 10. $56 + 9$: add 10 gives 66. Then take 1 back gives 65. So $56 + 9 = 65$. Adding 99? Add 100, then take 1 back. $56 + 99$: $56 + 100 = 156$, then minus 1 = 155.

Part C — Subtraction tricks

■ **Say: "Number 45" — Take Away in Parts**

Take away the big parts first, then the small parts. 56 minus 23: first take away the 20. 56 minus $20 = 36$. Now take away the 3: 36 minus $3 = 33$. So 56 minus $23 = 33$.

■ **Say: "Number 46" — Count Back**

When you take away 1, 2 or 3, do not work it out — count down. 49 minus 3: 48, 47, 46. Three little steps back. 49 minus $3 = 46$.

■ **Say: "Number 47" — Count Up**

If two numbers are close, count UP from the smaller one. 63 minus 58: start at 58... 59, 60, 61, 62, 63. That is 5 steps, so 63 minus $58 = 5$. Counting up is far easier than counting back here.

■ **Say: "Number 48" — Bridge Down**

Bridge Down means stepping onto the ten first. 52 minus 7: how many gets you down to 50? That is 2. Split the 7 into $2 + 5$. 52 minus $2 = 50$, then 50 minus $5 = 45$. So 52 minus $7 = 45$.

■ **Say: "Number 49" — Think Addition**

Every subtraction is an addition in disguise. For 15 minus 8, ask: $8 + ? = 15$. $8 + 7 = 15$, so 15 minus $8 = 7$. Use the addition facts you already know.

■ **Say: "Number 50" — Round and Adjust**

Round the number you are taking away to something tidy. 73 minus 38: round 38 up to 40. 73 minus $40 = 33$. You took away 2 too many, so add 2 back: $33 + 2 = 35$. So 73 minus $38 = 35$.

■ **Say: "Number 51" — Same Difference**

The difference does not change if you shift BOTH numbers the same way. 63 minus 28: add 2 to each gives 65 minus 30. 65 minus $30 = 35$. So 63 minus $28 = 35$. Much tidier!

■ **Say: "Number 52" — Subtract from 100**

To subtract from 100, find the pair that makes 100. 100 minus 63: $63 + 37 = 100$.

So 100 minus 63 = 37. Tip: if the number does NOT end in 0, the ones digits add to 10 and the tens digits add to 9. Ends in 0 (like 100 minus 40)? Just use tens bonds: $4 + 6 = 10$, so the answer is 60.

■ **Say: "Number 53" — Subtract 9 and 99**

Taking away 9 is nearly taking away 10. 64 minus 9: take away 10 gives 54. Give 1 back gives 55. So 64 minus 9 = 55. Taking away 99? Take away 100, then give 1 back.

Part D — Division tricks

■ **Say: "Number 54" — Times Tables Backwards**

Division is the OPPOSITE of multiplication. 56 divided by 7 is just asking: "7 times what makes 56?" 7 times 8 = 56, so 56 divided by 7 = 8. If your times tables are sharp, division is instant — every division fact is just a times-table fact flipped.

■ **Say: "Number 55" — $\div$ 10, 100, 1000**

To divide by 10, take ONE zero off the end: 470 divided by 10 = 47. To divide by 100, take TWO zeros off: 4700 divided by 100 = 47. To divide by 1000, take THREE zeros off: 47000 divided by 1000 = 47. Tip: it is really the decimal point moving LEFT — one place for each zero in the divisor.

■ **Say: "Number 56" — The 2, 5, 10 Test**

You can tell at a glance if a number divides by 2, 5 or 10: , divided by 2: last digit is 0, 2, 4, 6 or 8 (even). , divided by 5: last digit is 0 or 5. , divided by 10: last digit is 0. So 346 divides by 2 (ends in 6). 345 divides by 5 (ends in 5). 340 divides by all three. No actual division needed — just peek at the last digit.

■ **Say: "Number 57" — Digit Sum for 3 and 9**

For 3: if the digits add to a multiple of 3, the number divides by 3. 4-7-1: $4 + 7 + 1 = 12$. 12 divided by 3 = 4, so YES — 471 divides by 3. For 9: if the digits add to a multiple of 9, the number divides by 9. 5-4-9: $5 + 4 + 9 = 18$. 18 divided by 9 = 2, so YES — 549 divides by 9. Trick within a trick: keep adding the digits of the digit-sum until you have one digit. If it is 9, the number divides by 9.

■ **Say: "Number 58" — $\div 4$ = Halve Twice**

Dividing by 4 is the same as halving TWICE. 96 divided by 4: half of 96 is 48. Half of 48 is 24. So 96 divided by 4 = 24. Halving is usually easier in your head than divided by 4 in one go. Works because $4 = 2 \text{ times } 2$.

■ **Say: "Number 59" — $\div 8$ = Halve Three Times**

Dividing by 8 is halving THREE times in a row. 120 divided by 8: half is 60. Half of 60 is 30. Half of 30 is 15. So 120 divided by 8 = 15. Three quick halves and you are done. Works because $8 = 2 \text{ times } 2 \text{ times } 2$.

■ **Say: "Number 60" — $\div 5$ = Double then $\div 10$**

Dividing by 5 is the same as doubling, then dividing by 10. 630 divided by 5: double 630 gives 1260. divided by 10 gives 126. So 630 divided by 5 = 126. Why? Because $1 \text{ over } 5 = 2 \text{ over } 10$. You can do it in either order — double first or divided by 10 first — pick whichever looks easier.

■ **Say: "Number 61" — Last Two for 4, Last Three for 8**

For divided by 4: look ONLY at the last two digits. If THEY divide by 4, the whole number does too. 7,316: last two are 16. 16 divided by 4 = 4, so 7,316 divides by 4. For divided by 8: look ONLY at the last three digits. If they divide by 8, the whole number does. 5,128: last three are 128. 128 divided by 8 = 16, so 5,128 divides by 8. Saves you from dividing the whole big number.

■ **Say: "Number 62" — Halve Both First**

If the dividend AND divisor are both even, halve them both first. 164 divided by 26: both even. Halve both gives 82 divided by 13. 82 divided by 13 = 6 remainder 4. The 6 — the whole-number answer — is the same. But careful: any REMAINDER got halved with everything else, so double it back: 164 divided by 26 = 6 remainder 8. You can keep halving as long as both are even. 320 divided by 16 gives 160 divided by 8 gives 80 divided by 4 gives 40 divided by 2 gives 20. All the same answer: 20.

■ **Say: "Number 63" — Split the Dividend**

Split the BIG number (the dividend) into chunks that each divide cleanly. 864 divided by 6: split 864 into $600 + 264$. 600 divided by 6 = 100. 264 divided by 6 = 44. Add the answers: $100 + 44 = 144$. So 864 divided by 6 = 144. Two easy divisions instead of one hard one.

■ **Say: "Number 64" — Front-End First**

Mental division is easier from LEFT to right. 256 divided by 8: ask first — about how many tens? 8 times 30 = 240, that fits. So the answer starts with 30-something. Now what is left: 256 minus 240 = 16. 16 divided by 8 = 2. Put them together: 30 + 2 = 32. So 256 divided by 8 = 32.

■ **Say: "Number 65" — Subtract Friendly Chunks**

Build the answer up by SUBTRACTING easy multiples of the divisor. 432 divided by 6: start big. 6 times 50 = 300. Take it off: 432 minus 300 = 132. 6 times 20 = 120. Take it off: 132 minus 120 = 12. 6 times 2 = 12. Take it off: 12 minus 12 = 0. Add the friendly multipliers: 50 + 20 + 2 = 72. So 432 divided by 6 = 72.

■ **Say: "Number 66" — Round to Estimate**

When the numbers are awkward, ROUND them to something tidier. 347 divided by 6: round 347 to 360 (a multiple of 6). 360 divided by 6 = 60. So the answer is about 60. That tells you the size of the answer — and stops silly mistakes. Always estimate before you start a big division — you will spot a wrong answer straight away.

■ **Say: "Number 67" — The 11 Test**

For divisibility by 11, ADD and SUBTRACT the digits alternately. 2-9-1-5: 2 minus 9 + 1 minus 5 = minus 11. minus 11 divided by 11 = minus 1, so YES — 2915 divided by 11 = 265. If the alternating sum is 0 or any multiple of 11, the number divides by 11. Try 1-3-2: 1 minus 3 + 2 = 0 gives YES, 132 divided by 11 = 12. Otherwise: remember 11s as 11 times 12 = 132, 11 times 13 = 143, 11 times 14 = 154...

■ **Say: "Number 68" — The 7 Trick**

For divisibility by 7: DOUBLE the last digit and SUBTRACT it from the rest. 364: last digit 4, double = 8. Rest = 36. 36 minus 8 = 28. 28 divided by 7 = 4, so YES. Then 364 divided by 7 = 52. If the test number is small enough, you can see straight away whether it divides by 7. For most everyday sevens, just lean on your 7 times table — it is the most-tested fact in this whole topic.

Part E — Fractions tricks

■ **Say: "Number 69" — Same Bottom → Add the Tops**

If two fractions have the SAME bottom number, leave the bottom alone and just add (or subtract) the tops. 2 over 7 + 3 over 7: bottoms match. Add the tops: $2 + 3 = 5$. Answer: 5 over 7. 5 over 9 minus 1 over 9: bottoms match. Subtract: $5 \text{ minus } 1 = 4$. Answer: 4 over 9. Everything else with fractions builds on this single rule.

■ **Say: "Number 70" — Magic Fraction Facts**

A handful of fraction facts make EVERYTHING faster. Memorise these: 1 over 2 = 0.5 = 50 per cent. 1 over 4 = 0.25 = 25 per cent. 3 over 4 = 0.75 = 75 per cent. 1 over 5 = 0.2 = 20 per cent. 2 over 5 = 0.4 = 40 per cent. 1 over 10 = 0.1 = 10 per cent. 1 over 3 is about 0.33 is about 33 per cent. 2 over 3 is about 0.67 is about 67 per cent. 1 over 8 = 0.125 = 12.5 per cent. Once you know these, you barely have to calculate.

■ **Say: "Number 71" — Fraction of a Number**

To find a fraction OF a number: DIVIDE by the bottom, then MULTIPLY by the top. 3 over 4 of 20: divide by 4 gives 5. Multiply by 3 gives 15. So 3 over 4 of 20 = 15. 2 over 3 of 18: divided by 3 = 6. times 2 = 12. So 2 over 3 of 18 = 12. Always do the divided by first — it keeps the numbers small.

■ **Say: "Number 72" — Simplify by Halving**

If the top AND bottom are both even, halve them both. Keep going until you cannot any more. 12 over 16: both even. Halve gives 6 over 8. Halve again gives 3 over 4. That is fully simplified. If both end in 0, you can save a step by slashing the zeros (see "Slash the Zeros"). Halving is the easiest place to start — try it before any other simplifying.

■ **Say: "Number 73" — Slash the Zeros**

If the top AND bottom both end in 0, you can SLASH a zero off each. 30 over 50: cross out the zeros gives 3 over 5. Done. 120 over 200: slash one zero from each gives 12 over 20. Slash again gives no, only the 20 still ends in a zero. Use halving instead: 12 over 20 gives 6 over 10 gives 3 over 5. Slashing zeros is just dividing top and bottom by 10. Same idea, much faster to write.

■ **Say: "Number 74" — Compare to One Half**

Halfway is when the TOP is exactly half the BOTTOM. 4 over 8 is exactly a half (because 4 is half of 8). 3 over 8 is LESS than a half (3 is less than half of 8). 5 over 8 is MORE than a half (5 is more than half of 8). This is the quickest way to compare a fraction to 1 over 2 — no maths needed, just a quick look.

■ **Say: "Number 75" — Cross-Multiply to Compare**

To compare two fractions, cross-multiply: top times OTHER bottom. 2 over 3 vs 5 over 8: 2 times 8 = 16. 5 times 3 = 15. 16 is bigger, and 16 came from 2 over 3 — so 2 over 3 is the bigger fraction. It works because you are really making a common denominator without writing it down. Whichever PRODUCT is biggest, that side wins.

■ **Say: "Number 76" — Cancel Before Multiplying**

When you multiply fractions, look for shared factors and SLASH them out FIRST. 3 over 4 times 8 over 9: the 3 and 9 share a 3 gives cancel to 1 and 3. The 4 and 8 share a 4 gives cancel to 1 and 2. You are left with 1 over 1 times 2 over 3 = 2 over 3. Cancelling FIRST keeps the numbers small. Way easier than multiplying first and simplifying after. Tip: cancel any TOP with any BOTTOM — diagonally, or within the same fraction. Never top-with-top or bottom-with-bottom.

■ **Say: "Number 77" — Keep, Change, Flip**

To divide by a fraction: KEEP the first, CHANGE divided by to times, FLIP the second. 1 over 2 divided by 1 over 4: keep 1 over 2. Change divided by to times. Flip 1 over 4 gives 4 over 1. Now multiply: 1 over 2 times 4 over 1 = 4 over 2 = 2. So 1 over 2 divided by 1 over 4 = 2. Why? Dividing by 1 over 4 is the same as asking "how many quarters fit?" — and multiplying by 4 gives the same answer. It always works: KEEP — CHANGE — FLIP.

■ **Say: "Number 78" — Butterfly Method**

For adding fractions with DIFFERENT bottoms, draw a butterfly: 2 over 3 + 1 over 4: multiply the wings — 2 times 4 = 8, and 1 times 3 = 3. Top of answer = 8 + 3 = 11. Multiply the bottoms for the new bottom: 3 times 4 = 12. So 2 over 3 + 1 over 4 = 11 over 12. No need to find a common denominator first — the butterfly does it for you.

■ **Say: "Number 79" — Spot the Easy Common Bottom**

Before reaching for the butterfly, check: is one bottom a MULTIPLE of the other?
1 over 4 + 3 over 8: 8 is a multiple of 4. So just change 1 over 4 to 2 over 8. Now both bottoms match: 2 over 8 + 3 over 8 = 5 over 8. Done. Always look for the easy common bottom first. Butterfly is the back-up plan.

■ **Say: "Number 80" — Fraction → Percent**

Per cent means "out of 100". So to turn a fraction into a percentage, make the bottom 100. 3 over 5: multiply top and bottom by 20 gives 60 over 100. That is 60 per cent. 7 over 20: multiply both by 5 gives 35 over 100. That is 35 per cent. If the bottom is not a factor of 100, divide top by bottom on paper to get the decimal, then times 100.

Part F — Percentages tricks

■ **Say: "Number 81" — 10% — Shift the Decimal**

10 per cent means "1 out of every 10" — divide by 10. Quickest way: move the decimal point ONE place to the LEFT. 10 per cent of 47 gives 4.7. 10 per cent of 360 gives 36. 10 per cent of 8 gives 0.8. This is the single most important percentage trick. Almost every other percentage builds from here.

■ **Say: "Number 82" — 1% — Shift TWICE**

1 per cent means "1 out of every 100" — divide by 100. Move the decimal point TWO places to the LEFT. 1 per cent of 300 gives 3. 1 per cent of 47 gives 0.47. 1 per cent of 1200 gives 12. Combine with 10 per cent and you can build any percentage at all.

■ **Say: "Number 83" — % ↔ Fraction Magic**

These are worth knowing by heart — they make per cent problems instant: 50 per cent = 1 over 2. 25 per cent = 1 over 4. 75 per cent = 3 over 4. 20 per cent = 1 over 5. 10 per cent = 1 over 10. 1 per cent = 1 over 100. 12.5 per cent = 1 over 8. 33 per cent is about 1 over 3. 66 per cent is about 2 over 3. "25 per cent off pounds 80" is the same as "divided by 4" gives pounds 20 off. Much faster than calculating.

■ **Say: "Number 84" — Build from 10, 5 and 1**

You can build ANY percentage from a few 10 per cents, 5 per cents and 1 per cents. 15 per cent of 60: 10 per cent is 6. 5 per cent is half of that gives 3. Add gives $6 + 3 = 9$. 23 per cent of 200: 10 per cent is 20. 20 per cent is 40. 1 per cent is 2. 3 per cent is 6. Add gives $40 + 6 = 46$. Always start with 10 per cent — then halve, double, or add 1 per cents to land on your number.

■ **Say: "Number 85" — Halve and Double**

Once you have 10 per cent, two quick moves give you a lot more: 5 per cent = HALVE the 10 per cent. 20 per cent = DOUBLE the 10 per cent. 2.5 per cent = HALVE the 5 per cent. 5 per cent of 80: 10 per cent is 8. Halve gives 4. 20 per cent of 80: 10 per cent is 8. Double gives 16. Most everyday percentages are reached in 1 or 2 halvings or doublings.

■ **Say: "Number 86" — Swap the Numbers**

A magical fact: x per cent of y is ALWAYS the same as y per cent of x. So 4 per cent of 50 = 50 per cent of 4 = 2. Way easier the second way. 12 per cent of 25 = 25 per cent of 12 = 3 (a quarter of 12). Whenever the percentage looks awkward, try SWAPPING — sometimes a horrible question becomes an easy one.

■ **Say: "Number 87" — "Is over Of"**

To work out what PERCENT one number is of another: (part divided by whole) times 100. 30 out of 40 questions right? $30 \text{ divided by } 40 = 0.75$. times 100 = 75 per cent. 12 out of 50? $12 \text{ divided by } 50 = 0.24$. times 100 = 24 per cent. Useful tip: think "IS over OF". "What percent IS 30 OF 40?" gives 30 over 40.

■ **Say: "Number 88" — Multiplier Method (+% or -%)**

To ADD a percentage, multiply by $1 + (\text{per cent as a decimal})$. +25 per cent gives times 1.25. +10 per cent gives times 1.1. +50 per cent gives times 1.5. To TAKE OFF a percentage, multiply by $1 \text{ minus } (\text{per cent as a decimal})$. minus 25 per cent gives times 0.75. minus 10 per cent gives times 0.9. minus 50 per cent gives times 0.5. Example: pounds 80 + 25 per cent gives $80 \text{ times } 1.25 = \text{pounds } 100$. pounds 80 minus 25 per cent gives $80 \text{ times } 0.75 = \text{pounds } 60$. One multiply, no add-on step. Fast.

■ **Say: "Number 89" — Chain Two %s with Multipliers**

For TWO percentage changes, multiply the two multipliers. +10 per cent then minus 15 per cent: times 1.10 times 0.85 = times 0.935. So the total change is a 6.5 per cent DROP — not 5 per cent as you might guess. A 20 per cent rise followed by a 20 per cent fall is times 1.2 times 0.8 = times 0.96 — a 4 per cent loss overall. Useful for sales, taxes, interest, anything that stacks.

■ **Say: "Number 90" — Reverse Percentages**

If you only know the answer AFTER a percentage change, you can work back.

After 50 per cent off, the sale price is pounds 45. What was the original? pounds 45 is 50 per cent of the original, so original = pounds 45 divided by 50 times 100 = pounds 90. Or just multiply by 2 (because 50 per cent = half). Rule: divide the new amount by the per cent LEFT (as a decimal). pounds 60 = 75 per cent left gives 60 divided by 0.75 = pounds 80 original.

■ **Say: "Number 91" — Tip Combos**

Restaurant tip maths is just three combinations: 15 per cent = 10 per cent + 5 per cent (half of the 10 per cent added on). 20 per cent = 10 per cent times 2. 18 per cent = 20 per cent minus 2 per cent (or just round up from 15 per cent). A pounds 45 bill: 10 per cent is pounds 4.50. 20 per cent is pounds 9. 15 per cent is pounds 4.50 + pounds 2.25 = pounds 6.75. Once you know 10 per cent, every tip is a few seconds of mental maths.

■ **Say: "Number 92" — Use a Fraction for Tidy %s**

Some percentages are MUCH faster as fractions. 60 per cent of 35: 60 per cent = 3 over 5. 3 over 5 of 35 = 21. 25 per cent of 48: 25 per cent = 1 over 4. 48 divided by 4 = 12. 75 per cent of 80: 75 per cent = 3 over 4. 80 divided by 4 times 3 = 60.

Whenever the percentage is 10, 20, 25, 50, 75 (and the number divides nicely), the fraction route is fastest.

THE END — thank you! Save the audio into OneDrive > Times tables > recordings and say the word.